Regression & Correlational Analysis

From Predictive Modeling to Economic Elasticities

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Abstract

This presentation provides a comprehensive guide to Correlation and Regression Analysis, bridging statistical theory with applied econometrics.

- **Foundations:** Historical context, correlation vs. causation, and matrix algebra.
- **Modeling:** Simple and Multiple Linear Regression (SLR & MLR) using automotive and clinical data.
- **Diagnostics:** Rigorous testing of regression assumptions (Normality, Homoscedasticity, Multicollinearity).
- **Econometric Applications:**
  - Modeling macroeconomic inflation (CPI) using **Marginal Elasticities**.
  - Analyzing microeconomic demand functions (price, income, and cross-price elasticities).
  - Estimating the Keynesian Consumption Function and the Marginal Propensity to Consume (MPC).

We emphasize the translation of statistical outputs into actionable economic, clinical, and business implications.

Session Objectives

- Distinguish between correlation and causation in observational data.
- Compute and interpret Pearson and Spearman correlation coefficients.
- Formulate, estimate, and interpret SLR and MLR models.
- Diagnose regression violations using formal statistical tests (Shapiro-Wilk, Breusch-Pagan, VIF).
- **Advanced:** Apply log-log transformations to estimate economic elasticities.
- Estimate microeconomic demand functions to derive own-price, income, and cross-price elasticities.
- Understand and estimate the Keynesian Marginal Propensity to Consume (MPC) using time-series data.

Part 1: Foundations of Regression

"The historical and theoretical underpinnings."

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Regression Analysis Defined

The Core Concept

A statistical technique used to examine the relationship between one or more **independent variables** (predictors) and a **dependent variable** (outcome).

Primary Use Cases

- **Prediction:** Forecasting continuous outcomes.
- **Inference:** Identifying important factors that influence the outcome.
- **Control:** Understanding the marginal effect of a single variable.

The Mathematical Form

For a simple linear model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where Y is the outcome, X is the predictor, β_0 is the intercept, β_1 is the slope, and ϵ is the error term.

History of Regression Analysis

- **Francis Galton (Late 1800s):** Coined the term "regression" while studying the heights of parents and their children. He observed that extreme heights in parents tended to "regress toward the mean" in their offspring.
- **Karl Pearson & Ronald Fisher (Early 20th Century):** Formalized the mathematical foundations of correlation and regression, integrating them into the broader framework of inferential statistics and experimental design.
- **Modern Era:** Regression has evolved from a purely statistical tool to a foundational algorithm in machine learning, econometrics, and biostatistics.

Correlation vs. Causation

The Golden Rule of Statistics

Correlation does not imply causation.

- **Correlation:** A statistical measure describing the degree to which two variables move together. It quantifies *association*.
- **Causation:** Implies that a change in one variable *directly produces* a change in another.
- **The Trap:** Two variables may be highly correlated due to:
  1. Direct causation (A causes B).
  2. Reverse causation (B causes A).
  3. A hidden confounding variable (C causes both A and B).
  4. Pure coincidence (Spurious correlation).

Transition

Part 2: Correlation Analysis

"Quantifying the strength of association."

The `mtcars` Dataset

- Extracted from the 1974 *Motor Trend US* magazine.
- Contains fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

Variable	Description	Type
`mpg`	Miles/(US) gallon	Continuous
`wt`	Weight (1000 lbs)	Continuous
`hp`	Gross horsepower	Continuous
`cyl`	Number of cylinders	Categorical/Ordinal
`am`	Transmission (0=auto, 1=manual)	Binary

Visualizing Relationships: Scatter Plots

The Prerequisite
Scatter plots are **essential** before running any regression or correlation analysis. They reveal the shape of the relationship and highlight potential outliers.

Observation: Weight vs. MPG

- Clear **negative linear trend**.
- As weight increases, fuel efficiency drops.
- No obvious non-linear patterns or severe outliers violating the linear assumption.

Seaborn `regplot`

Using `sns.regplot` in Python automatically overlays the OLS regression line and confidence intervals on the scatter plot, providing an immediate visual check of the model fit.

Anscombe's Quartet

Four entirely different datasets can have the *exact same* mean, variance, correlation, and regression line. Only visualization reveals the true data structure.

Pearson Correlation Analysis

- Measures the **linear** relationship between two continuous variables.
- Ranges from -1 (perfect negative) to +1 (perfect positive).

Variables	**Pearson's** r	**t-statistic**	**p-value**
`wt` & `mpg`	**-0.8676**	-9.559	1.29×10^{-10}

- **Interpretation:** A strong negative linear correlation. Heavier cars consistently achieve fewer miles per gallon.
- **Significance:** $p < 0.001$. We reject the null hypothesis of zero correlation.

Spearman Rank Correlation

- A non-parametric alternative to Pearson's r.
- Measures the **monotonic** relationship (based on ranked values) rather than strictly linear.
- **When to use:** When data is ordinal, contains severe outliers, or violates the normality assumption.

Variables	**Spearman's ρ**	**Statistic**	**p-value**
`wt & mpg`	**-0.8864**	10292	1.49×10^{-11}

- The Spearman correlation (-0.886) is slightly stronger than Pearson's, suggesting a highly consistent monotonic decrease in MPG as weight increases, robust to any extreme outliers.

Correlation & Covariance Matrices

- Examines pairwise relationships between multiple continuous variables simultaneously.

	mpg	**cyl**	**disp**	**hp**	**wt**
mpg	1.00	-0.85	-0.85	-0.78	-0.87
cyl	-0.85	1.00	0.90	0.83	0.78
disp	-0.85	0.90	1.00	0.79	0.89
hp	-0.78	0.83	0.79	1.00	0.66
wt	-0.87	0.78	0.89	0.66	1.00

- **Insight:** `wt` and `disp` are highly correlated (0.89). Including both in a multiple regression model may cause multicollinearity.

Part 3: Simple Linear Regression (SLR)

"Modeling continuous outcomes with a single predictor."

SLR Concept & Equation

The Simple Linear Regression Model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Y: Dependent variable (Outcome, e.g., `mpg`).
- X: Independent variable (Predictor, e.g., `wt`).
- β_0: Y-intercept (Value of Y when $X = 0$).
- β_1: Slope coefficient (Change in Y for a 1-unit increase in X).
- ϵ: Error term (Residuals, assumed to be normally distributed with mean 0).

SLR Model Output (`mtcars`)

Predictor	Estimate	Std. Error	t value	Pr $>$t
(Intercept)	37.285	1.878	19.86	$< 2e-16$ ***
`wt`	-5.344	0.559	-9.56	$1.29e-10$ ***

Interpretation

- **Slope ($\beta_1 = -5.344$):** For every 1,000 lbs increase in weight, MPG decreases by 5.344 units.
- **Significance:** $p < 0.001$. We reject H_0. Weight is a highly significant predictor of fuel efficiency.

SLR R-Squared & Confidence Intervals

- **Multiple R-squared (0.753):** 75.3% of the variance in MPG is explained by the vehicle's weight.
- **F-statistic:** 91.38 on 1 and 30 DF, $p < 0.001$.

Predictor	**2.5% CI**	**97.5% CI**
(Intercept)	33.45	41.12
`wt`	-6.49	-4.20

- We are 95% confident that the true effect of weight on MPG lies between -6.49 and -4.20. Because the interval **does not include zero**, it confirms statistical significance.

Prediction & Confidence vs. Prediction Intervals

Forecasting a New Car
Let's predict the MPG for a car weighing **3.5** (3,500 lbs).

- **Predicted Mean MPG:** 18.58

The Two Intervals

- **Confidence Interval (Mean):** $[17.43, 19.73]$
- **Prediction Interval (Observation):** $[12.25, 24.90]$

What's the difference?

- **Confidence Interval:** Estimates where the *average* MPG of *all* 3500 lb cars will fall. It is narrower.
- **Prediction Interval:** Estimates where a *single, specific* 3500 lb car's MPG will fall. It is wider because it accounts for individual variance (ϵ).

SLR Real-World Application: Hypertension Data

- **Dataset:** Clinical hypertension data ($N \approx 5000$).
- **Challenge:** Missing values in `BMI`, `hgb`, `HIV` status.
- **Strategy:** Complete case analysis (drop NA) $\rightarrow$ Model `SBP` $\sim$ `BMI`.

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	92.841	1.872	49.59	$< 2e-16$ ***
`BMI`	0.857	0.086	10.01	$< 2e-16$ ***

- **Equation:** $\text{SBP} = 92.84 + 0.857(\text{BMI})$
- **Interpretation:** For every 1 unit increase in BMI, SBP increases by 0.857 mmHg.
- **R-squared:** 0.041 (Only 4.1% of variance explained. Blood pressure is highly multifactorial!).

Part 4: Multiple Linear Regression (MLR)

"Controlling for confounders and improving model fit."

MLR Concept & Hypotheses

The Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

- **Scenario:** Predicting `mpg` using both `wt` (weight) and `hp` (horsepower).
- **Hypotheses for Predictors:**
  - **H:** $\beta_i = 0$ (The predictor has no effect on Y, *holding other variables constant*).
  - **H:** $\beta_i \neq 0$ (The predictor has a significant effect).

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MLR Model Output (`mtcars`)

Predictor	Estimate	Std. Error	t value	Pr >t
(Intercept)	37.227	1.599	23.28	$< 2e-16$ ***
`wt`	-3.878	0.633	-6.13	$1.12e-06$ ***
`hp`	-0.032	0.009	-3.52	0.00145 **

Interpretation

- **Weight (`wt`):** A 1,000 lb increase decreases MPG by 3.88 units, *holding horsepower constant*.
- **Horsepower (`hp`):** A 1 unit increase in HP decreases MPG by 0.032 units, *holding weight constant*.
- Both predictors are statistically significant ($p < 0.05$).

MLR Model Fit (R-Squared)

- **Multiple R-squared:** 0.827
- **Adjusted R-squared:** 0.815

Interpretation

By adding `hp` to the model, the explained variance in MPG increased from 75.3% (SLR) to 82.7% (MLR).

The **Adjusted R-squared** (0.815) is the preferred metric for MLR, as it penalizes the addition of irrelevant predictors. It indicates that 81.5% of the variability in fuel efficiency is explained by weight and horsepower combined.

Part 5: Regression Diagnostics

"Ensuring the validity of our statistical inferences."

The Core Assumptions of Linear Regression

- Ordinary Least Squares (OLS) regression relies on strict assumptions. If violated, coefficients may be biased, or p-values invalid.

The "LINE" Assumptions

1. **L**inearity: The relationship between X and Y is linear.
2. **I**ndependence: Observations (residuals) are independent (no autocorrelation).
3. **N**ormality: The residuals are normally distributed.
4. **E**qual Variance (Homoscedasticity): The variance of residual errors is constant across all levels of X.

Diagnostic 1: Multicollinearity

- Occurs when independent variables are highly correlated with each other.
- **Consequence:** Inflates standard errors, making coefficients unstable and statistically insignificant.
- **Detection:** Variance Inflation Factor (VIF).

Predictor	VIF
`wt`	1.77
`hp`	1.77

- **Rule of Thumb:** VIF > 5 (or 10) indicates severe multicollinearity. Here, VIF is low; no action needed.

Diagnostic 2: Influential Observations

- Some data points may exert disproportionate leverage on the regression line.
- **Detection:** Cook's Distance.
- **Threshold:** Observations with Cook's Distance $> \frac{4}{N}$ are considered highly influential.

Action Taken

In the `mtcars` MLR model, 4 influential cars were identified: *Chrysler Imperial, Fiat 128, Toyota Corolla, Maserati Bora*.

Best Practice: Do not blindly delete outliers. Investigate them first. If they are data entry errors, remove them. If they are genuine extreme cases, consider robust regression techniques.

Refitting Without Influential Observations

The Experiment
We drop the 4 flagged cars and refit the MLR model on the remaining 28 observations.

Before vs. After

- **R-squared:** $0.827 \rightarrow \mathbf{0.892}$
- **Adj. R-squared:** $0.815 \rightarrow \mathbf{0.883}$
- **wt Coef:** $-3.878 \rightarrow \mathbf{-3.752}$
- **hp Coef:** $-0.032 \rightarrow \mathbf{-0.033}$

The Result

Removing the extremes significantly improved the model's explanatory power. The coefficients stabilized, and the standard errors decreased, confirming that the original model was being distorted by those specific outliers.

Diagnostic Improvement

The Omnibus test for normality improved drastically to $p = 0.879$, indicating the residuals are now much closer to a normal distribution.

Diagnostic 3: Normality of Residuals

The Tests

- **Shapiro-Wilk:** Best for small to medium samples ($N < 5000$). Tests H_0: Data is normal.
- **Anderson-Darling:** More sensitive to the tails of the distribution.

Our Results (Original MLR)

- Shapiro-Wilk $W = 0.928$, $p = 0.034$
- Anderson-Darling Statistic $= 0.666$

Since $p < 0.05$, we **reject** the null hypothesis. The residuals are *not* perfectly normal.

Visual Confirmation

The Q-Q plot shows points deviating from the red diagonal line at the tails, and the histogram reveals slight right-skewness and kurtosis.

Is this fatal?

With $N = 32$, minor deviations from normality are common and often tolerable. However, it justifies our decision to investigate and remove the influential outliers, which subsequently improved the normality profile.

Diagnostic 4: Homoscedasticity (Breusch-Pagan)

The Concept
Homoscedasticity means "same scatter." The spread of the errors should be uniform. If the variance changes (e.g., errors get larger as X increases), it's called **heteroscedasticity**.

The Breusch-Pagan Test
H_0: Residuals are homoscedastic (constant variance).
H_1: Heteroscedasticity is present.

Our Results
BP Statistic $= 0.88$, p-value $= 0.64$.

Interpretation

Since $p = 0.64 > 0.05$, we **fail to reject** the null hypothesis. We conclude that the variance of the errors is constant. The homoscedasticity assumption is **satisfied**.

Why it matters

If heteroscedasticity is present, the OLS coefficients remain unbiased, but their **standard errors are incorrect**, rendering t-tests and p-values unreliable. The fix is to use **Robust Standard Errors** (HC3).

Diagnostic 5: Independence (Durbin-Watson)

The Concept
Errors should not be correlated with each other. This is primarily a concern in **time-series data** (autocorrelation), where today's error might be related to yesterday's error.

The Durbin-Watson Statistic
Ranges from 0 to 4.

- **DW $\approx$ 2:** No autocorrelation.
- **DW < 1.5:** Positive autocorrelation.
- **DW > 2.5:** Negative autocorrelation.

Our Results

- `mtcars` MLR: DW $= 1.36$ (Slight positive autocorrelation, but acceptable for cross-sectional data).
- Macroeconomics MLR: DW $= 0.41$ (**Severe positive autocorrelation**).

The Time-Series Trap

The macroeconomic dataset is a time series. A DW of 0.41 violates the independence assumption. Standard OLS is invalid here; we must use time-series specific models (ARIMA, GLS) or include lagged variables.

Part 6: Macroeconomic Modeling

"Predicting inflation using macroeconomic indicators."

Macroeconomic Model Context

- **Dataset:** Annual macroeconomic indicators (`economics` dataset).
- **Target Variable:** Personal Savings Rate (`psavert`).
- **Predictors:** Personal consumption (`pce`), unemployment (`unemploy`), population (`pop`), and median unemployment duration (`uempmed`).

Standard MLR Output

- **R-squared:** 0.841
- **F-statistic:** 752.1 ($p < 0.001$)
- **Durbin-Watson:** 0.410 (Severe autocorrelation detected)

The Log-Log Model Specification

The Elasticity Model

$$\ln(Y) = \beta_0 + \beta_1 \ln(X_1) + \beta_2 \ln(X_2) + \epsilon$$

- In this specification, β_1 represents the **elasticity** of Y with respect to X_1.
- **Definition:** A 1% increase in X_1 leads to a β_1% change in Y, holding other variables constant.
- This is ideal for macroeconomic variables that grow exponentially over time, as it linearizes the relationship and stabilizes variance.

Log-Log Model Output & Interpretation

Predictor	Estimate	Std. Error	t value	p-value
(Intercept)	11.011	6.795	1.621	0.106
ln(pce)	**-0.361**	0.084	-4.291	< 0.001 ***
ln(unemploy)	**0.409**	0.065	6.305	< 0.001 ***
ln(pop)	-0.801	0.572	-1.401	0.162
ln(uempmed)	**0.162**	0.066	2.448	0.015 **

- **R-squared:** 0.802
- **Economic Insight:** A 1% increase in the number of unemployed individuals is associated with a 0.409% *increase* in the savings rate (precautionary saving behavior).

Part 7: Applied Econometrics

"Estimating elasticities and the Marginal Propensity to Consume."

CPI & Elasticities: Exchange Rates & Interest Rates

The Log-Log Equation
Modeling Consumer Price Index (CPI) against Exchange Rates and Lending Interest Rates.

$$\ln(CPI) = -0.704 + 1.515 \ln(ExchRate) - 0.552 \ln(LendInt)$$

Model Fit

- **R-squared:** 0.935
- **F-statistic:** 164.8 ($p < 0.001$)

Elasticity Interpretation

- **Exchange Rate:** 1.515. A 1% depreciation in the currency leads to a 1.515% increase in CPI (imported inflation).
- **Interest Rates:** −0.552. A 1% increase in lending rates decreases CPI by 0.552% (cooling demand).

Diagnostics

Unlike the previous macro model, this one shows excellent normality (Omnibus $p = 0.952$), making these elasticity estimates highly reliable for policy analysis.

Demand Analysis: Double-Log Model

The Setup
Modeling the quantity demanded of a good based on its price, consumer income, and the prices of related goods.

Model Performance

- **R-squared:** 0.910
- **All predictors significant** at $p < 0.001$.

Elasticity Interpretations

- **Price ($E_p = -1.139$):** Demand is **elastic**. A 1% price increase drops demand by 1.14%.
- **Income ($E_Y = 0.753$):** Good is a **normal necessity**.
- **Substitute ($E_{Sub} = 0.537$):** Positive sign confirms they are **substitutes**.
- **Complement ($E_{Comp} = -0.407$):** Negative sign confirms they are **complements**.

Keynesian Consumption Function: Estimating the MPC

The Standard SLR

$\text{Consumption} = 11.37 + 0.898 \times \text{Income}$

Interpretation: The Marginal Propensity to Consume (MPC) is 0.898. For every \$1 increase in income, consumption rises by 89.8 cents.

The Long-Run Model (No Intercept)

We force the regression line through the origin (`Consumption ~ Income - 1`).

$$\text{Consumption} = 0.905 \times \text{Income}$$

Why force the intercept to zero?

In the *long run*, economic theory dictates that if income is zero, consumption must be zero (assuming no savings/debt). The standard model's intercept (\$11.37) represents short-run autonomous consumption (survival spending).

Model Fit

Both models have an R-squared near 1.000. The choice between them depends entirely on the **economic theory** being tested (short-run vs. long-run behavior), not just statistical fit.

Q&A and Thank You!

Questions?

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